MSC 201 **99372**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A box has 3 red and 7 blue tokens; 5 are drawn without replacement. Why is 4 red impossible?	A. Half the population is drawn. B. The sample contains 5 tokens. C. Only 3 red tokens exist. D. Red probability must be zero. 10 pts
2. The success proportion is 0.35. What is the failure proportion?	A. 0.175. B. 0.35. C. 1.35. D. 0.65. 10 pts
3. To estimate a population mean with unknown population SD using a sample SD, which standard method fits?	A. A one-sample t interval. B. A known-SD z interval treating s as known population SD. C. A prediction interval for a single future observation. D. A proportion interval regardless of variable type. 10 pts
4. The 95th percentile of a sample-mean distribution is 14. What does it describe?	A. Every sample mean is 14. B. 95% of all individuals must be at most 14. C. The unknown mean has become 14. D. About 95% of repeated sample means are at most 14. 10 pts
5. A systematic sample starts at ID 3 and takes every tenth ID. Which IDs appear first?	A. 3, 10, 20. B. 3, 6, 9. C. 3, 13, 23. D. 10, 20, 30. 10 pts
6. A normal model has mean 50 and SD 2. Which interval contains about 99.7%?	A. 48 to 52. B. 50 to 56. C. 46 to 54. D. 44 to 56. 10 pts
7. A dataset has mean 10. One new observation equal to 10 is added. What happens to the mean?	A. It rises. B. It falls. C. It becomes zero. D. It stays 10. 10 pts

Test continues on the other side of the sheet.

MSC 201 **99372**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. X is uniform from 2 to 10. What can be said about P(X>12)?	A. It is zero. B. It is 1/2. C. It is larger than P(X>8). D. It is the density height at 12. 10 pts
9. An inspector checks a fixed batch of independent items with the same defect chance. Which variable fits a binomial model?	A. The exact mass of the first item. B. The attempt on which the first defect is found. C. The number of defective items found. D. The average time needed to check an item. 10 pts
10. For two trials with positive chance of two failures, how do exactly one failure and at least one failure compare?	A. At least one has greater probability. B. Exactly one has greater probability. C. They must be equal. D. The two events are complements. 10 pts

99372

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3. A.

10 pts

4.

D.

10 pts

5.

C.

10 pts

6.

D.

10 pts

7.

D.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **48020**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A sample is drawn without replacement from a box containing marked and unmarked cards. When must the sample contain a marked card?	A. Only if every card in the box is marked. B. When the sample size exceeds the number of unmarked cards. C. Whenever one marked card is present in the box. D. Whenever the sample size is less than the number of unmarked cards. 10 pts
2. A question asks for the mass exceeded by only 10% of packages. What units should the answer have?	A. Probability units only. B. A percentage of sample size. C. Mass units. D. No units because it is a z-score automatically. 10 pts
3. Daily print totals are shown in a histogram instead of a time-series graph. What information is lost?	A. Which total occurred on which date. B. The approximate distribution of totals. C. The existence of high totals. D. The number of daily observations. 10 pts
4. Two distinct terminal paths describe exactly one head in two tosses. How is their combined probability found?	A. Subtract the smaller from the larger. B. Multiply the two path probabilities. C. Add the two path probabilities. D. Ignore one because both describe one head. 10 pts
5. Two services have mean wait 10 minutes, but SDs 1 and 6 minutes. What does the larger SD suggest?	A. The mean must actually be 16 minutes. B. Every wait is six minutes longer. C. Waits vary more around the same mean. D. The median must be 6 minutes. 10 pts
6. A supplied sampling model gives P(mean<=12)=0.90. What is P(mean>12)?	A. 0.10. B. 1.90. C. 0.90. D. 0.50. 10 pts
7. Which question targets a population proportion?	A. What fraction of jobs finish on time? B. What is the mean duration of jobs? C. What is the SD of file sizes? D. What is the median number of pages? 10 pts

Test continues on the other side of the sheet.

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. For a continuous duration, how do P(X<6) and P(X<=6) compare?	A. The second is always twice the first. B. They differ by the density height at 6. C. The first is always zero. D. They are equal. 10 pts
9. X counts trials through first success. What does X>3 require?	A. Failures on the first three trials. B. Exactly three successes. C. At least one success in the first three trials. D. A success on trial 3. 10 pts
10. The plus-four estimate adds two successes and two failures. With 8 successes out of 40, what adjusted fraction is used?	A. 8/44. B. 10/40. C. 10/44. D. 12/44. 10 pts

48020

TEST PAGE 1

1. B.

10 pts

2. C.

10 pts

3. A.

10 pts

4. C.

10 pts

5. C.

10 pts

6. A.

10 pts

7. A.

10 pts

TEST PAGE 2

8. D.

10 pts

9. A.

10 pts

10. C.

10 pts

MSC 201 **46675**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Two mixed populations have the same size, but one has a larger share of faulty cables. Both are sampled randomly without replacement at the same sample size. How do their expected faulty counts compare?	A. The expected counts depend only on whether order is recorded. B. The expected count is higher for the population with a larger faulty share. C. The expected count is lower for the population with a larger faulty share. D. The expected counts are equal because the sample sizes agree. 10 pts
2. Half of tablet users use laptops, and half of non-tablet users use laptops. What does this suggest in these data?	A. The events are mutually exclusive. B. Every student uses exactly one device. C. Laptop use does not depend on tablet-use status. D. Every student uses both devices. 10 pts
3. Compare 4, 5, 6 with 1, 5, 9. Which statement is correct?	A. They have equal means, but the second has more spread. B. The first has more spread. C. Equal means imply equal spread. D. The second has a larger mean. 10 pts
4. A mean interval is 10 to 14 minutes. Which report matches its target?	A. The median is guaranteed to equal 12. B. The procedure estimates the population mean duration between 10 and 14 minutes. C. Every future duration will be between 10 and 14. D. Exactly 95 of the next 100 durations will lie there. 10 pts
5. Independent sample size quadruples with population SD fixed. What happens to SE of the mean?	A. It quarters. B. It doubles. C. It stays unchanged. D. It halves. 10 pts

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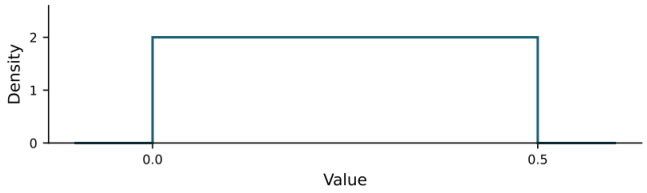
MSC 201 **46675**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

6. A density is flat at height 2 over a half-unit interval and zero elsewhere. Is height 2 alone a problem? 	A. No; total area may be any positive number. B. Yes; every density height must be at most 1. C. Yes; its probability is 2 everywhere. D. No; its total area is 1, and height is not probability. 10 pts
7. A timer reports durations to the nearest second. What follows about the underlying duration?	A. It may vary continuously despite rounded records. B. It cannot be less than one second. C. It must change only in whole seconds. D. It is a categorical label. 10 pts
8. A strongly skewed variable is converted to z-scores. Does it become normal?	A. No; standardization shifts and rescales without removing skew. B. Yes; every z-score distribution is standard normal. C. Yes; subtracting the mean removes the tail. D. No; z-scores cannot be negative. 10 pts
9. One student counts failed attempts before the first success. Another counts all attempts through the first success. How are their variables related?	A. Neither count can be known when the first success occurs. B. The first includes one extra trial. C. The second includes one extra trial, the success itself. D. Both counts are always equal. 10 pts
10. A report gives 30 successes among 120 observations. Which is the estimate of population success proportion?	A. 90. B. 30. C. 120. D. 0.25. 10 pts

46675

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

D.

10 pts

TEST PAGE 2

6.

D.

10 pts

7. A.

10 pts

8. A.

10 pts

9.

C.

10 pts

10.

D.

10 pts

MSC 201 **71512**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A count variable has expected value 0.4. What does that mean?	A. The model is invalid. B. Its long-run average can be 0.4 despite integer observations. C. A single session must contain 0.4 events. D. Exactly 40% of sessions must contain one event. 10 pts
2. Given that a U(0,12) wait exceeds 4, what shape describes the total wait's conditional density?	A. Flat over 0 to 12 unchanged. B. A point at 4. C. Flat over 4 to 12. D. Normal over all real numbers. 10 pts
3. The ordered data are 2, 4, 4, 7, 20. What is the median?	A. 7. B. 20. C. 4. D. 6. 10 pts
4. Two normal curves share SD but have different means. What is equal?	A. Their probabilities below every fixed cutoff. B. Their widths and shapes. C. Their 90th-percentile values. D. Their centers. 10 pts
5. The number of independent items rises from 25 to 100. What happens to expected sum and SD of sum?	A. Expected sum quadruples; SD doubles. B. Expected sum stays fixed; SD halves. C. Both quadruple. D. Both halve. 10 pts
6. A 95% confidence interval estimates a population use rate. Does 95% mean that 95% of the population uses the service?	A. No; it means 95 people were surveyed. B. Yes; confidence equals the use rate. C. No; 95% describes the interval procedure's coverage. D. Yes; all sample proportions equal 0.95. 10 pts
7. Why report a confidence interval instead of only a sample mean?	A. It lists every individual observation. B. It guarantees the true mean is the midpoint. C. It removes sampling bias. D. It expresses uncertainty in estimating the population mean. 10 pts

Test continues on the other side of the sheet.

MSC 201 **71512**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A first branch has probability 0.4; its next conditional branch is 0.5. What is the path probability?	A. 0.9. B. 0.1. C. 0.8. D. 0.2. 10 pts
9. Which wording is most neutral for a parking survey?	A. Do you support the unfair parking fee? B. Do you oppose the helpful parking fee? C. Do you support the proposed parking fee? D. Why do you dislike the parking fee? 10 pts
10. The sample is an unordered set of cables. Why count combinations rather than different orders of the same set?	A. Reordering the same selected cables does not create a new sample set. B. Selection order determines whether a cable is faulty. C. Every sample set has only one cable. D. Combinations assume replacement. 10 pts

71512

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5. A.

10 pts

6.

C.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **52622**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. X is uniform from 2 to 10. To find P(0<X<5), which portion contributes area?	A. 0 to 2. B. 2 to 5. C. 0 to 5. D. 5 to 10. 10 pts
2. Notebook colors are blue, blue, black, red. Which statement meaningfully summarizes a typical category?	A. The mean color is 2. B. The standard deviation of color names is 1. C. Blue is the most common color. D. The median is 2.5 colors. 10 pts
3. A fixed-size sample is randomly selected from a batch of known good and faulty bulbs. Each selected bulb is set aside. Which model fits the number of good bulbs selected?	A. Geometric. B. Continuous normal. C. Binomial. D. Hypergeometric. 10 pts
4. A small random independent sample comes from a normal population. Is n<30 alone a reason to reject a t interval?	A. Yes; t requires at least 30 in every case. B. Yes; normality is irrelevant below 30. C. No; it must use a proportion interval. D. No; small normal samples can support the t method. 10 pts
5. Individual masses have SD 20 g; means of 25 independent masses have SD 4 g. Which spread applies to one package?	A. 20 g. B. 4 g. C. 16 g. D. 0.8 g. 10 pts
6. Which event is the complement of at least one failure in three trials?	A. Exactly one failure. B. All three failures. C. At least two failures. D. No failures. 10 pts
7. A dataset has mean 8. One observation below 8 is removed. What happens to the mean of the remaining values?	A. It always stays 8. B. It decreases. C. It becomes the removed value. D. It increases. 10 pts

Test continues on the other side of the sheet.

MSC 201 **52622**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A normal model's mean increases, with SD fixed. At a fixed deadline, what happens to the probability of finishing before it?	A. It becomes exactly 0.5. B. It increases. C. It stays unchanged. D. It decreases. 10 pts
9. A bar graph of counts 98 and 100 starts its vertical axis at 97. What impression might it exaggerate?	A. The size of the difference between the counts. B. The number of categories. C. The fact that 98 is below 100. D. The total number of observations. 10 pts
10. All values and their mean and SD are converted from meters to centimeters. What happens to z-scores?	A. They stay unchanged. B. They multiply by 100. C. They divide by 100. D. They all become positive. 10 pts

52622

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5. A.

10 pts

6.

D.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

D.

10 pts

9. A.

10 pts

10. A.

10 pts

MSC 201 **87377**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Students choosing tutoring score higher. Why is a causal claim uncertain?	A. Motivation may affect both tutoring attendance and scores. B. Any score difference proves a tutoring effect. C. A larger score difference removes all confounding. D. Tutoring can never help students. 10 pts
2. X counts broken cables in independent checks. Can broken be called success in a binomial model?	A. Only if no cables are broken. B. Only if more than half are broken. C. Yes; success means the counted outcome. D. No; success must be desirable. 10 pts
3. The success chance improves after each failed attempt because the process learns. Why is a geometric model questionable?	A. There is a stopping rule. B. Failures are counted. C. The first success occurs eventually. D. The success probability is not constant. 10 pts
4. Someone says n=30 guarantees normal sample means for every population. What is the problem?	A. Thirty is too large for any sample. B. Skewness and extreme observations affect how large n needs to be. C. Sample means can never approach normality. D. Only populations with mean zero qualify. 10 pts
5. A service meets its deadline with probability 0.80. What is the probability of missing it?	A. 1.80. B. 0.80. C. 0.40. D. 0.20. 10 pts
6. A rate is stated per hour but a cutoff is inserted as a number of minutes. What must be done before using the tail formula?	A. Round the probability to an integer. B. Express rate and cutoff in compatible time units. C. Multiply both numbers by the sample size. D. Replace the cutoff with the mean. 10 pts
7. For a comparison holding the observed proportion and confidence fixed, n quadruples. What happens to estimated margin?	A. It stays unchanged. B. It halves. C. It doubles. D. It quarters. 10 pts

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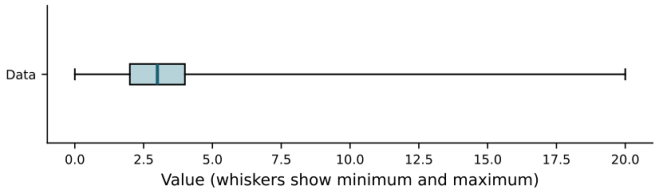
MSC 201 **87377**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A basic box plot reports minimum 0, Q1=2, median 3, Q3=4, maximum 20. Which comparison is supported? 	A. The segment from Q3 to the maximum spans a wider numerical range than the segment from the minimum to Q1. B. The upper segment must contain three times as many observations. C. The lower segment contains no observations. D. The plot establishes the exact mean. 10 pts
9. A fitted normal model predicts 2% above a cutoff, while a large observed dataset has 18% above it. What should be investigated?	A. Whether probabilities above 2% are impossible. B. Whether observations should be changed to fit the model. C. Only the number of decimal places. D. Whether the model fits the observed tail. 10 pts
10. A mean interval is 10 to 14 minutes. Which report matches its target?	A. Every future duration will be between 10 and 14. B. The procedure estimates the population mean duration between 10 and 14 minutes. C. Exactly 95 of the next 100 durations will lie there. D. The median is guaranteed to equal 12. 10 pts

87377

TEST PAGE 1

1. A.

10 pts

2. C.

10 pts

3. D.

10 pts

4. B.

10 pts

5. D.

10 pts

6. B.

10 pts

7. B.

10 pts

TEST PAGE 2

8. A.

10 pts

9. D.

10 pts

10. B.

10 pts

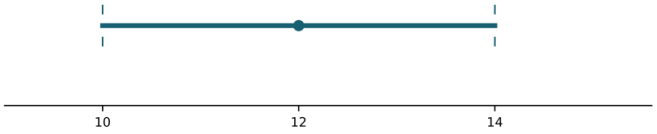
MSC 201 **83436**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Normal models have the same mean. For a fixed cutoff below that mean, which has the larger lower-tail probability?	A. The model with the larger SD. B. The one with more recorded observations regardless of SD. C. They must have equal tails. D. The model with the smaller SD. 10 pts
2. Sample mean is 12 and margin of error is 2. What interval results? 	A. 2 to 12. B. 10 to 14. C. 6 to 24. D. 12 to 14. 10 pts
3. A score is reported at approximately the 90th percentile. What kind of information does that give?	A. A score of exactly 90 points. B. That 90% of exam answers were correct. C. The class mean being 90. D. Its position relative to other scores. 10 pts
4. A box has 3 red and 7 blue tokens; 5 are drawn without replacement. Why is 4 red impossible?	A. The sample contains 5 tokens. B. Red probability must be zero. C. Only 3 red tokens exist. D. Half the population is drawn. 10 pts
5. Students rate a service while its manager watches. Compared with anonymous responses, what might happen?	A. Students may avoid negative ratings. B. Every rating must become negative. C. The manager's presence guarantees accuracy. D. Sample size automatically doubles. 10 pts
6. Which quantity naturally has discrete possible values?	A. An unrounded measured length. B. An unrounded file-transfer duration. C. The number of failed files. D. The exact mass of a package. 10 pts

Test continues on the other side of the sheet.

MSC 201 **83436**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. An exact continuous duration has P(X=4)=0. Can a timer report 4 seconds?	A. Only if the density equals zero everywhere. B. No; 4 seconds can never be recorded. C. Only if the model becomes discrete before rounding. D. Yes; rounding collects a positive-width interval of exact values. 10 pts
8. Score A is 10 above its mean with SD 5; score B is 12 above its mean with SD 12. Which is relatively farther above its group mean?	A. They are equally far. B. B. C. The larger raw score difference always wins. D. A. 10 pts
9. Two samples have the same observed mean above a claimed population mean and the same population SD, but sizes 25 and 100. Which mean is farther above in SE units?	A. The larger sample changes the claimed mean. B. The size-25 mean. C. They are equally far. D. The size-100 mean. 10 pts
10. A symmetric central normal interval contains 90%. How much area lies in each outside tail?	A. 45%. B. 5%. C. 10%. D. 90%. 10 pts

83436

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. D.

10 pts

4. C.

10 pts

5. A.

10 pts

6. C.

10 pts

TEST PAGE 2

7. D.

10 pts

8. D.

10 pts

9. D.

10 pts

10. B.

10 pts

MSC 201 **90669**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A probability table totals 1 but contains one entry of -0.1. What follows?	A. It is valid if the mean is positive. B. Negative probabilities simply indicate rare events. C. It is valid because only the total matters. D. It is invalid because probabilities cannot be negative. 10 pts
2. A well-run randomized experiment finds better results for an app among its volunteers. Which claim needs extra caution?	A. The same effect applies to all students everywhere. B. Random assignment helped compare the formats. C. The observed volunteers had different average results. D. The study provides evidence about an app effect in this setting. 10 pts
3. A population has mean 50. Where are unbiased random-sample means centered?	A. At the population SD. B. At zero. C. At 50 divided by sample size. D. At 50. 10 pts
4. Two reasonable quartile methods give upper outlier fences of 18 and 20 for the same dataset. One observation is 35. Which conclusion agrees with both methods?	A. The observation is not flagged under either method. B. The observation must be a recording error. C. Only the method with fence 18 flags the observation. D. The observation is flagged as unusually high under both methods. 10 pts
5. Why use planning proportion 0.5 when no prior estimate is available?	A. It guarantees half the population has the trait. B. It maximizes p(1-p), giving a conservative required sample size. C. It makes the sample unbiased automatically. D. It minimizes all margins to zero. 10 pts

Test continues on the other side of the sheet.

MSC 201 **90669**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

6. Two mixed populations have the same size, but one has a larger share of faulty cables. Both are sampled randomly without replacement at the same sample size. How do their expected faulty counts compare?	A. The expected count is higher for the population with a larger faulty share. B. The expected counts are equal because the sample sizes agree. C. The expected counts depend only on whether order is recorded. D. The expected count is lower for the population with a larger faulty share. 10 pts
7. A team redraws randomly from a mixed deck, replacing and remixing each card. It stops at the first heart. Which model describes the number of draws?	A. Hypergeometric. B. Continuous uniform. C. Geometric. D. Binomial. 10 pts
8. A question asks for the mass exceeded by only 10% of packages. What units should the answer have?	A. Probability units only. B. No units because it is a z-score automatically. C. Mass units. D. A percentage of sample size. 10 pts
9. A known-SD mean interval currently has margin 1. To target margin 0.5 at the same confidence, how should n change?	A. Halve n. B. Add 1 to n. C. Multiply n by 4. D. Multiply n by 2. 10 pts
10. A wait is uniform from 0 to 12 minutes. What is P(3<X<6)?	A. 6/12. B. 3/12. C. 3/6. D. 9/12. 10 pts

90669

TEST PAGE 1

1.

D.

10 pts

2. A.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5.

B.

10 pts

TEST PAGE 2

6. A.

10 pts

7.

C.

10 pts

8.

C.

10 pts

9.

C.

10 pts

10.

B.

10 pts

MSC 201 **49906**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A student treats z=2 as a probability of 2. What is wrong?	A. A normal model cannot have positive z-scores. B. A z-score can never exceed 1. C. A z-score must be a percentage. D. A standardized distance is not an area probability. 10 pts
2. What puts an upper limit on the number of marked cards in a sample drawn without replacement?	A. Both the sample size and the number of marked cards in the box. B. The previous sample's marked-card count. C. Only the number of unmarked cards in the box. D. Only the probability of a marked card on the first draw. 10 pts
3. An exponential model has mean wait 4 minutes. What is its SD?	A. 16 minutes. B. 0 minutes. C. 2 minutes. D. 4 minutes. 10 pts
4. Fifty readings taken seconds apart from one fluctuating machine are treated as independent. What should be checked?	A. Whether 50 is an integer. B. Whether readings have units. C. Dependence between nearby readings. D. Whether the mean can be calculated. 10 pts
5. A sample has 8 observations. Its sum of squared deviations is already supplied. Which divisor gives sample variance?	A. 64. B. 7. C. 8. D. The sample mean. 10 pts
6. Which change best addresses a poll that includes only frequent library users?	A. Broaden selection to include nonusers. B. Collect ten times as many frequent-user responses. C. Report more decimal places. D. Shuffle the existing responses. 10 pts

Test continues on the other side of the sheet.

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A sample variance is 9 minutes squared. What is its standard deviation?	A. 3 minutes squared. B. 81 minutes. C. 3 minutes. D. 9 minutes. 10 pts
8. A proportion estimate is 0.40 with margin 0.05. What interval is this in percentages?	A. 39.95% to 40.05%. B. 40% to 50%. C. 5% to 40%. D. 35% to 45%. 10 pts
9. Normal models have the same mean. For a fixed cutoff below that mean, which has the larger lower-tail probability?	A. The model with the smaller SD. B. The one with more recorded observations regardless of SD. C. They must have equal tails. D. The model with the larger SD. 10 pts
10. A first branch has probability 0.4; its next conditional branch is 0.5. What is the path probability?	A. 0.2. B. 0.8. C. 0.1. D. 0.9. 10 pts

49906

TEST PAGE 1

1.

D.

10 pts

2. A.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5.

B.

10 pts

6. A.

10 pts

TEST PAGE 2

7.

C.

10 pts

8.

D.

10 pts

9.

D.

10 pts

10. A.

10 pts

MSC 201 **56756**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A supplied probability table lists P(X=0),...,P(X=8). To find P(X<=2), which entries are needed?	A. Only the entry for 2. B. The entries for 0, 1, and 2. C. The entries for 3 through 8. D. The expected value. 10 pts
2. Sample SD is 6 minutes and n=9; sqrt(9)=3. What estimated SE is used?	A. 6 minutes. B. 3 minutes squared. C. 18 minutes. D. 2 minutes. 10 pts
3. To halve SE, how must an independent sample's size change?	A. Multiply n by 2. B. Add 4 regardless of its current size. C. Divide n by 2. D. Multiply n by 4. 10 pts
4. An exponential model has mean wait 4 minutes. What is its SD?	A. 0 minutes. B. 4 minutes. C. 2 minutes. D. 16 minutes. 10 pts
5. A table has positive frequency at exactly 2 jobs. How do counts for fewer than 2 and at most 2 compare?	A. Fewer than 2 has the larger count. B. Neither can be found from exact frequencies. C. They must be equal. D. At most 2 has the larger count. 10 pts
6. Among 200 recorded jobs, 30 were late. What does 30/200 describe?	A. A guarantee that the next job is late. B. The exact long-run probability under every possible condition. C. The number of late jobs that must occur tomorrow. D. The observed late-job relative frequency. 10 pts
7. For a standard normal variable, compare P(Z<-1.4) and P(Z>1.4).	A. Both equal 0.5. B. The left tail is larger. C. The right tail is larger. D. They are equal. 10 pts

Test continues on the other side of the sheet.

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Two reasonable quartile methods give upper outlier fences of 18 and 20 for the same dataset. One observation is 35. Which conclusion agrees with both methods?	A. Only the method with fence 18 flags the observation. B. The observation is not flagged under either method. C. The observation must be a recording error. D. The observation is flagged as unusually high under both methods. 10 pts
9. Most files are small, with a few extremely large files. Which better describes a typical file's size?	A. The mean student ID. B. The total size alone. C. The median. D. The maximum. 10 pts
10. Only 5% exceed cutoff k. Which percentile is k?	A. 5th. B. 95th. C. 50th. D. 100th. 10 pts

56756

TEST PAGE 1

1.

B.

10 pts

2.

D.

10 pts

3.

D.

10 pts

4.

B.

10 pts

5.

D.

10 pts

6.

D.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

C.

10 pts

10.

B.

10 pts

MSC 201 **13010**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Compare U(0,12) and U(20,32). What is true about their standard deviations?	A. They are equal. B. Both standard deviations are zero. C. The second is larger because its mean is larger. D. The first is larger because it includes zero. 10 pts
2. A symmetric confidence interval is 8 to 12. What point estimate is at its center?	A. 4. B. 8. C. 12. D. 10. 10 pts
3. A game has negative expected net gain. Can a player still win money on one play?	A. No; every outcome must lose money. B. Only after the expected value changes. C. Yes, if positive-gain outcomes have positive probability. D. No; the largest prize must be negative. 10 pts
4. For 2, 3, 3, 4, 18, how does the mean compare with the median?	A. They are equal. B. The mean is smaller. C. No comparison is possible without a calculator. D. The mean is larger. 10 pts
5. What equation describes a continuous model's 90th percentile k?	A. P(X>k)=0.90. B. P(X<=k)=0.90. C. P(X=k)=0.90. D. The mean equals 0.90k. 10 pts
6. Datasets share min=0 and max=100, but have different IQRs. What follows?	A. Equal ranges do not imply equal middle-half spread. B. Their medians must be identical. C. Their observations must be identical. D. Their standard deviations must be identical. 10 pts

Test continues on the other side of the sheet.

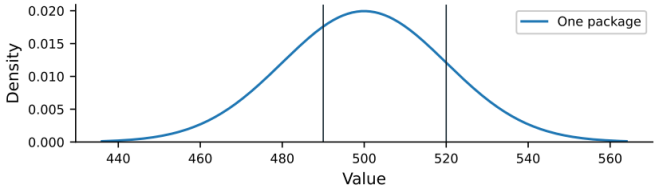
MSC 201 **13010**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A package is accepted if 490<mass<520. Which part of its normal curve should be shaded? 	A. Everything above 520. B. Both outside tails. C. Only the area between 490 and 520. D. Only the point 500. 10 pts
8. A researcher deletes genuine results that disagree with the desired conclusion. What is the problem?	A. A smaller dataset guarantees less bias. B. The report no longer fairly represents the collected evidence. C. Any unusual observation is automatically an error. D. Deleting inconvenient values always improves accuracy. 10 pts
9. Tokens are replaced and mixed after each draw. Compared with leaving them out, what does replacement preserve?	A. A decreasing total number of tokens. B. The same token being selected every time. C. The original color probabilities on each independent draw. D. A guarantee that colors alternate. 10 pts
10. For n=20, a total exceeding 240 is equivalent to which mean event?	A. Mean exceeding 240. B. Mean below 12. C. Mean exceeding 12. D. Mean exceeding 20. 10 pts

13010

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. C.

10 pts

4. D.

10 pts

5. B.

10 pts

6. A.

10 pts

TEST PAGE 2

7. C.

10 pts

8. B.

10 pts

9. C.

10 pts

10. C.

10 pts

MSC 201 **62066**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. X has mean 50, SD 10. Which standard-normal event corresponds to X<60?	A. Z<6. B. Z<1. C. Z<-1. D. Z>1. 10 pts
2. A dataset has mean 8. One observation below 8 is removed. What happens to the mean of the remaining values?	A. It becomes the removed value. B. It decreases. C. It always stays 8. D. It increases. 10 pts
3. The numbers of pages in six jobs are 1, 2, 2, 4, 4, 5. Which statement about the mode is correct?	A. Both 2 and 4 are modes. B. The mode is 3, halfway between 2 and 4. C. Only 2 is a mode. D. Only 4 is a mode. 10 pts
4. For one normal model, P(X<40)=0.20 and P(X<60)=0.85. What is P(40<X<60)?	A. 0.20. B. 0.65. C. 1.05. D. 0.85. 10 pts
5. Students choosing tutoring score higher. Why is a causal claim uncertain?	A. A larger score difference removes all confounding. B. Any score difference proves a tutoring effect. C. Tutoring can never help students. D. Motivation may affect both tutoring attendance and scores. 10 pts
6. A model has one blank probability entry and lists every possible value. What does the blank need to account for?	A. The average of the listed probabilities. B. All probability not already assigned to the other values. C. Half the total probability, regardless of the other entries. D. The probability of the most likely listed value. 10 pts

Test continues on the other side of the sheet.

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A sample of 200 is drawn without replacement from 500. Does it meet the course's at-most-10% independence guideline?	A. Yes; every random sample meets the guideline. B. No; it is 40% of the population. C. No; samples must contain fewer than 10 observations. D. Yes; 200 is larger than 30. 10 pts
8. Durations are rounded to the nearest whole second. Ignoring tie conventions at boundaries, which exact interval reports as 4?	A. 4 to 5 seconds. B. Only the exact point 4. C. 0 to 4 seconds. D. 3.5 to 4.5 seconds. 10 pts
9. At fixed n and confidence, known population SD doubles. What happens to interval width?	A. It stays unchanged. B. It doubles. C. It quadruples. D. It halves. 10 pts
10. A payout variable is measured in dollars. What units belong to its variance?	A. Dollars squared. B. Dollars. C. No units. D. Percent. 10 pts

62066

TEST PAGE 1

1.

B.

10 pts

2.

D.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

D.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

D.

10 pts

9.

B.

10 pts

10. A.

10 pts

MSC 201 **20970**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Which question targets a population proportion?	A. What is the SD of file sizes? B. What is the mean duration of jobs? C. What is the median number of pages? D. What fraction of jobs finish on time? 10 pts
2. A sample of 6 has strong skewness and an extreme outlier. What is sensible about a t interval?	A. It is automatically reliable because t has heavy tails. B. The outlier must be removed without investigation. C. Its reliability deserves caution. D. Six observations guarantee a normal sampling distribution. 10 pts
3. An event is described as drawing a heart or a king. Does the king of hearts qualify?	A. Yes; the statistical OR includes the overlap. B. No; OR always excludes overlap. C. Only if hearts and kings are independent. D. Only if the card is drawn twice. 10 pts
4. Four jobs have a mean of 5 pages. What is the total number of pages?	A. 1.25. B. 5. C. 9. D. 20. 10 pts
5. Two binomial settings have the same fixed number of trials. One has a greater chance of success on each trial. What changes?	A. The expected number of successes is greater in that setting. B. Every run in that setting yields more successes. C. The maximum possible number of successes is greater in that setting. D. Both settings have the same expected number of successes. 10 pts
6. About 99.7% lies within 3 SD in a normal model. About what percent lies outside on both sides combined?	A. 0.3%. B. 3%. C. 99.7%. D. 0.15%. 10 pts

Test continues on the other side of the sheet.

MSC 201 **20970**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A game charges a fixed entry fee and sometimes awards a prize. What distinguishes net gain from the prize amount?	A. Net gain accounts for the fee paid to play. B. Net gain equals the largest possible prize. C. Net gain ignores outcomes with no prize. D. The fee changes the chance of each prize. 10 pts
8. Independent observations come from an exactly normal population. Is n>=30 necessary for their mean to be normal?	A. Yes; normal means are impossible below 30. B. No; the mean instead becomes uniform. C. Yes; individual normality disappears in small samples. D. No; their mean is normal for any positive sample size. 10 pts
9. A proposed density dips below zero over an interval. What is wrong?	A. Every valid density must be symmetric. B. Its maximum must equal 1. C. It would assign negative area to that interval. D. A density must be flat everywhere. 10 pts
10. Four mutually exclusive, exhaustive categories have reported proportions 0.20, 0.30, 0.40, and 0.25. What needs checking?	A. Proportions must be whole numbers. B. Each category must have proportion 0.25. C. No category may exceed 0.20. D. Their total exceeds 1. 10 pts

20970

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3. A.

10 pts

4.

D.

10 pts

5. A.

10 pts

6. A.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

D.

10 pts

9.

C.

10 pts

10.

D.

10 pts

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Which model assigns equal probability to all equal-length subintervals within its support?	A. Every continuous model. B. An exponential model. C. A continuous uniform model. D. A normal model. 10 pts
2. A report gives 30 successes among 120 observations. Which is the estimate of population success proportion?	A. 90. B. 120. C. 0.25. D. 30. 10 pts
3. Both endpoints of a uniform model increase by 4. What happens to its mean and SD?	A. SD rises by 4; mean stays unchanged. B. Both rise by 4. C. Mean rises by 4; SD stays unchanged. D. Both stay unchanged. 10 pts
4. A public dataset includes names beside individual survey answers. Which change best protects privacy while preserving group summaries?	A. Replace every answer with the overall mean. B. Keep names and omit the study methods. C. Remove identifying details before sharing. D. Publish only the names. 10 pts
5. On a die, A is even and B is greater than 3. Which results satisfy at least one condition?	A. 2, 4, 5, 6. B. All six results. C. 4, 6 only. D. 1, 3 only. 10 pts
6. Which design permits an exact binomial count model with independent randomized draws?	A. Replacing and remixing each drawn token. B. Drawing without replacement until the box is empty. C. Always stopping at first red. D. Drawing most of a small population without replacement. 10 pts
7. Most waits are short, but a few extend far toward large values. What determines the skew direction?	A. Whether the variable uses minutes. B. The number of categories. C. The side with the tallest bar alone. D. The long tail toward large values. 10 pts

Test continues on the other side of the sheet.

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Four jobs have a mean of 5 pages. What is the total number of pages?	A. 1.25. B. 20. C. 9. D. 5. 10 pts
9. A normal model's mean rises by 3 with SD fixed. What happens to its 90th percentile?	A. It rises by 90%. B. It stays unchanged. C. It rises by 3. D. It falls by 3. 10 pts
10. A histogram is made from 500 means, each based on 20 observations. How many values are in that histogram?	A. 520. B. 500. C. 20. D. 10,000. 10 pts

22124

TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4.

C.

10 pts

5. A.

10 pts

6. A.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

B.

10 pts

9.

C.

10 pts

10.

B.

10 pts

MSC 201 **77619**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Students rate a service while its manager watches. Compared with anonymous responses, what might happen?	A. Students may avoid negative ratings. B. The manager's presence guarantees accuracy. C. Sample size automatically doubles. D. Every rating must become negative. 10 pts
2. The wait until the next request is modeled as exponential with rate 4 per hour. What is its mean in minutes?	A. 4 minutes. B. 60 minutes. C. 240 minutes. D. 15 minutes. 10 pts
3. A sample variance is 9 minutes squared. What is its standard deviation?	A. 9 minutes. B. 3 minutes squared. C. 3 minutes. D. 81 minutes. 10 pts
4. Among 20 tablet users, 12 also use laptops. What is P(laptop given tablet) in this sample?	A. 12 divided by all college students. B. 12/20. C. 20/12. D. 8/20. 10 pts
5. A binomial experiment has many trials, but successes are so rare that only a few are expected. Why might a normal approximation be poor?	A. The trials must be dependent. B. The count is automatically symmetric because there are many trials. C. The expected number of successes is too small. D. The expected number of failures must also be small. 10 pts
6. All values and their mean and SD are converted from meters to centimeters. What happens to z-scores?	A. They multiply by 100. B. They divide by 100. C. They stay unchanged. D. They all become positive. 10 pts
7. A narrow interval uses only the fastest successful jobs to estimate the mean for all jobs. What is its main limitation?	A. Biased selection can make the interval misleading despite narrow width. B. Adding decimal places removes selection bias. C. Its narrow width guarantees representativeness. D. The mean cannot describe time. 10 pts

Test continues on the other side of the sheet.

MSC 201 **77619**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Six tokens are drawn without replacement from a small mixed box. What challenges an exact binomial model?	A. The number of draws is fixed. B. Draws change the composition and are dependent. C. There are only two colors. D. The count is an integer. 10 pts
9. 40 of 100 sampled jobs succeed. What is the sample success proportion?	A. 2.5. B. 0.40. C. 40. D. 0.60. 10 pts
10. X counts trials through first success. What does X>3 require?	A. At least one success in the first three trials. B. Failures on the first three trials. C. A success on trial 3. D. Exactly three successes. 10 pts

77619

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. C.

10 pts

4. B.

10 pts

5. C.

10 pts

6. C.

10 pts

7. A.

10 pts

TEST PAGE 2

8. B.

10 pts

9. B.

10 pts

10. B.

10 pts

MSC 201 **57190**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. In many sessions, 20% have no failures. How might this inform a model?	A. Set every count probability to 0.20. B. Infer that every fifth session must have no failures. C. Use 0.20 as an empirical estimate for P(X=0). D. Declare zero failures impossible. 10 pts
2. A sample has 2 successes and 98 failures. Does a large total alone justify the ordinary normal-approximation proportion interval?	A. Yes; n=100 guarantees every condition. B. Yes; failures are numerous. C. No; proportions cannot be estimated at all. D. No; the observed success count is too sparse. 10 pts
3. A uniform interval centered at 10 doubles its width while keeping center 10. What happens?	A. Both double. B. Both stay unchanged. C. Mean doubles; SD stays unchanged. D. Mean stays 10; SD doubles. 10 pts
4. One game always pays its expected payout. Another has the same expected payout but offers several possible payouts. How do they differ? 	A. Only the second game has positive spread. B. Their spreads must match because their means match. C. Only the first game has positive spread. D. Both games must have zero spread. 10 pts
5. A bell-shaped distribution has mean 20 and SD 2. Approximately which interval contains 95%?	A. 16 to 24. B. 0 to 20. C. 20 to 24. D. 18 to 22. 10 pts
6. Ten independent trials each have success probability 0.3. X counts successes. What makes a binomial model appropriate?	A. Every trial must succeed. B. The expected count is an integer. C. Success probability exceeds 0.5. D. Fixed trial count, two outcomes, independence, and constant success chance. 10 pts

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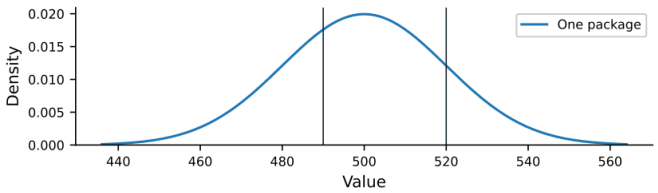
MSC 201 **57190**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. To estimate a population mean with unknown population SD using a sample SD, which standard method fits?	A. A known-SD z interval treating s as known population SD. B. A proportion interval regardless of variable type. C. A prediction interval for a single future observation. D. A one-sample t interval. 10 pts
8. Which area of the approximating normal curve represents exactly one specified integer count?	A. The area between the midpoints separating that integer from its two neighbors. B. The entire area to the right of the count. C. The whole area between the two neighboring integers. D. The height of the curve at the count. 10 pts
9. A package is accepted if 490<mass<520. Which part of its normal curve should be shaded? 	A. Everything above 520. B. Only the point 500. C. Both outside tails. D. Only the area between 490 and 520. 10 pts
10. Exact wait times are replaced by short, medium, and long categories. What information is lost?	A. Whether any observations exist. B. The number of observations. C. The ordering of the three categories. D. The exact differences between wait times. 10 pts

57190

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

D.

10 pts

4. A.

10 pts

5. A.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

D.

10 pts

8. A.

10 pts

9.

D.

10 pts

10.

D.

10 pts

MSC 201 **79208**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Two distinct terminal paths describe exactly one head in two tosses. How is their combined probability found?	A. Ignore one because both describe one head. B. Add the two path probabilities. C. Subtract the smaller from the larger. D. Multiply the two path probabilities. 10 pts
2. To estimate a population mean with unknown population SD using a sample SD, which standard method fits?	A. A known-SD z interval treating s as known population SD. B. A proportion interval regardless of variable type. C. A one-sample t interval. D. A prediction interval for a single future observation. 10 pts
3. A random sample of 80 enrolled students reports 30% support for later hours. Which conclusion is justified?	A. Exactly 30% of all students support later hours. B. Exactly 24 students in the entire college support later hours. C. 30% describes the sample and estimates population support. D. The population percentage must exceed 30%. 10 pts
4. A box has 3 red and 7 blue tokens; 5 are drawn without replacement. Why is 4 red impossible?	A. The sample contains 5 tokens. B. Red probability must be zero. C. Only 3 red tokens exist. D. Half the population is drawn. 10 pts
5. A report gives 30 successes among 120 observations. Which is the estimate of population success proportion?	A. 120. B. 0.25. C. 30. D. 90. 10 pts
6. SE is 4 at n=25. What is SE at n=100, with the same population SD and independence?	A. 1. B. 8. C. 4. D. 2. 10 pts

Test continues on the other side of the sheet.

MSC 201 **79208**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A payout variable is measured in dollars. What units belong to its variance?	A. Dollars squared. B. Dollars. C. No units. D. Percent. 10 pts
8. A continuous model has F(9)=0.80. What percent exceeds 9?	A. 9%. B. 20%. C. 80%. D. 100%. 10 pts
9. A proposed normal model gives substantial probability to negative durations. What is the concern?	A. It assigns appreciable probability to impossible values. B. Its mean must be zero. C. Normal distributions cannot model any positive variable. D. Every duration must be negative. 10 pts
10. A sample variance is 9 minutes squared. What is its standard deviation?	A. 3 minutes. B. 3 minutes squared. C. 9 minutes. D. 81 minutes. 10 pts

79208

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4.

C.

10 pts

5.

B.

10 pts

6.

D.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

B.

10 pts

9. A.

10 pts

10. A.

10 pts

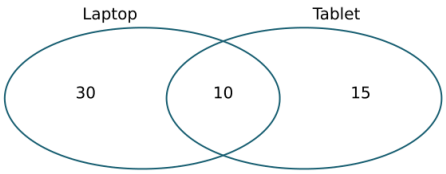
MSC 201 **26992**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Using the reported quartiles, group A has Q1=3, Q3=9; group B has Q1=10, Q3=14. Which has greater IQR?	A. They are equal. B. Group B. C. Group A. D. The larger median must determine the IQR. 10 pts
2. A teacher draws cards without replacement until the first marked card. Why is the draw number generally neither binomial nor geometric?	A. A success has to be a desirable outcome. B. Stopping at the first success always makes the count binomial. C. The trial count is not fixed, and the chance can change between draws. D. The draw number is continuous. 10 pts
3. For a comparison holding the observed proportion and confidence fixed, n quadruples. What happens to estimated margin?	A. It halves. B. It doubles. C. It quarters. D. It stays unchanged. 10 pts
4. After a specific interval is computed, what is correct about its relation to a fixed population mean?	A. The endpoints must equal the smallest and largest observation. B. The fixed mean changes to fit it with probability 95%. C. It either contains that mean or does not; 95% describes the procedure. D. The mean occupies 95% of the interval. 10 pts
5. 40 students use laptops, 25 use tablets, and 10 use both. Why subtract 10 when counting either device? 	A. Both-device users must be excluded entirely. B. Subtraction is required because the sample is random. C. Adding the first two counts includes the overlap twice. D. The overlap does not use either device. 10 pts
6. Arrival rates sharply rise at lunchtime. What is questionable about a single exponential waiting model over the whole day?	A. Nonnegative waiting times. B. A constant rate across the whole period. C. Allowing a long wait. D. Recording waits in minutes. 10 pts

Test continues on the other side of the sheet.

MSC 201 **26992**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A box has 3 red and 7 blue tokens; 5 are drawn without replacement. Why is 4 red impossible?	A. Red probability must be zero. B. Half the population is drawn. C. The sample contains 5 tokens. D. Only 3 red tokens exist. 10 pts
8. A college wants each academic program represented in a survey. Which plan best achieves that directly?	A. Survey only the largest program. B. Invite only students who previously complained. C. Randomly sample students within every program. D. Select the first students entering one room. 10 pts
9. A proposed normal model gives substantial probability to negative durations. What is the concern?	A. It assigns appreciable probability to impossible values. B. Its mean must be zero. C. Every duration must be negative. D. Normal distributions cannot model any positive variable. 10 pts
10. A supplied normal distribution of sample means gives cumulative probabilities 0.15 at 8 and 0.85 at 12. What is the probability between 8 and 12?	A. 0.15. B. 1.00. C. 0.85. D. 0.70. 10 pts

26992

TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3. A.

10 pts

4.

C.

10 pts

5.

C.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

C.

10 pts

9. A.

10 pts

10.

D.

10 pts

MSC 201 **96167**

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Circle one response per question in the right column.

1. The SD of a normal model increases while the mean stays fixed. What happens to its peak?	A. The center shifts right. B. It becomes lower as the curve spreads out. C. The total area rises above 1. D. It becomes higher and narrower. 10 pts
2. All ten observations equal 5, so their median is 5. Are exactly half strictly below the median?	A. No; none are strictly below it. B. Yes; every median splits strict inequalities equally. C. Yes; exactly five are below 5. D. No; a median cannot exist with ties. 10 pts
3. Both endpoints of a uniform model increase by 4. What happens to its mean and SD?	A. Both rise by 4. B. SD rises by 4; mean stays unchanged. C. Both stay unchanged. D. Mean rises by 4; SD stays unchanged. 10 pts
4. X counts the successes. Which values satisfy fewer than 3 successes?	A. 0, 1, 2, 3. B. 0, 1, 2. C. 1, 2, 3 only. D. 3 and above. 10 pts
5. The 95th percentile of a sample-mean distribution is 14. What does it describe?	A. About 95% of repeated sample means are at most 14. B. The unknown mean has become 14. C. 95% of all individuals must be at most 14. D. Every sample mean is 14. 10 pts
6. Using the same sample and method, confidence rises from 90% to 99%. What happens to interval width?	A. It increases. B. It decreases. C. It stays unchanged. D. It becomes zero. 10 pts

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7. One student counts failed attempts before the first success. Another counts all attempts through the first success. How are their variables related?	A. The first includes one extra trial. B. Both counts are always equal. C. The second includes one extra trial, the success itself. D. Neither count can be known when the first success occurs. 10 pts
8. A probability interval is widened to include its original interval plus a region of positive density. What happens to its probability?	A. It exceeds 1 automatically. B. It decreases. C. It stays unchanged. D. It increases. 10 pts
9. A bar graph of counts 98 and 100 starts its vertical axis at 97. What impression might it exaggerate?	A. The fact that 98 is below 100. B. The size of the difference between the counts. C. The total number of observations. D. The number of categories. 10 pts
10. For 2, 3, 3, 4, 18, how does the mean compare with the median?	A. They are equal. B. No comparison is possible without a calculator. C. The mean is larger. D. The mean is smaller. 10 pts

96167

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

D.

10 pts

4.

B.

10 pts

5. A.

10 pts

6. A.

10 pts

TEST PAGE 2

7.

C.

10 pts

8.

D.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **25750**

Diagnostic Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

1. Population SD is 12 and n=36, with independent observations. sqrt(36)=6. What is the mean's SE?	A. 6. B. 2. C. 72. D. 12. 10 pts
2. The sample is an unordered set of cables. Why count combinations rather than different orders of the same set?	A. Every sample set has only one cable. B. Reordering the same selected cables does not create a new sample set. C. Combinations assume replacement. D. Selection order determines whether a cable is faulty. 10 pts
3. The success proportion is 0.35. What is the failure proportion?	A. 0.175. B. 1.35. C. 0.65. D. 0.35. 10 pts
4. All values and their mean and SD are converted from meters to centimeters. What happens to z-scores?	A. They stay unchanged. B. They divide by 100. C. They multiply by 100. D. They all become positive. 10 pts
5. Which event is the complement of at least one failure in three trials?	A. All three failures. B. Exactly one failure. C. At least two failures. D. No failures. 10 pts
6. X is uniform from 0 to 12. Which symmetric interval contains the middle 50%?	A. 5 to 7. B. 0 to 6. C. 3 to 9. D. 6 to 12. 10 pts
7. A box plot has median 6. Can it reveal exactly how many observations equal 5?	A. No; the summary does not retain those exact counts. B. Yes; none equal 5. C. Yes; half equal 5. D. Yes; exactly one quarter equal 5. 10 pts

Test continues on the other side of the sheet.

MSC 201 **25750**

Diagnostic Exam

Name: _____ Date: _____

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Circle one response per question in the right column.

8. A sample-size formula returns 62.2 observations. Which planned integer size meets that minimum?	A. 61. B. 63. C. 60. D. 62. 10 pts
9. A college wants each academic program represented in a survey. Which plan best achieves that directly?	A. Survey only the largest program. B. Randomly sample students within every program. C. Invite only students who previously complained. D. Select the first students entering one room. 10 pts
10. A game charges a fixed entry fee and sometimes awards a prize. What distinguishes net gain from the prize amount?	A. Net gain equals the largest possible prize. B. Net gain accounts for the fee paid to play. C. Net gain ignores outcomes with no prize. D. The fee changes the chance of each prize. 10 pts

25750

TEST PAGE 1

1. B.

10 pts

2. B.

10 pts

3. C.

10 pts

4. A.

10 pts

5. D.

10 pts

6. C.

10 pts

7. A.

10 pts

TEST PAGE 2

8. B.

10 pts

9. B.

10 pts

10. B.

10 pts